PLANT SPECIES IDENTIFICATION FROM VEGETATION PLOT IMAGES

A PROJECT REPORT

Submitted by

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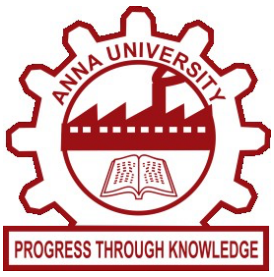
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OCTOBER 2026

BONAFIDE CERTIFICATE

Certified that this Report titled "**PLANT SPECIES IDENTIFICATION FROM VEGETATION PLOT IMAGES**" is the bonafide work of **SANTHIYA SHREE J (241801249) and SHALINI PUNITHAN (241801258)**, who carried out the project work under the supervision of the project guide. The report describes the design and implementation of a big-data analytics workflow for studying player and match behavior in online multiplayer games. Guide and institutional details should be verified before final submission.

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ACKNOWLEDGEMENT

Initially I thank the Almighty for being with us through every walk of my life and showering his blessings through the endeavor to put forth this report. My sincere thanks to our Chairman **Mr. S. MEGANATHAN, M.E., F.I.E.,** and our Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, M.E., Ph.D.,** for providing me with the requisite infrastructure and sincere endeavoring educating me in their premier institution. My sincere thanks to **Dr. S.N. MURUGESAN M.E., Ph.D.,** our beloved Principal for his kind support and facilities provided to complete our work in time. I express my sincere thanks to **Dr. J M Gnanasekar M.E., Ph.D.,** Head of the Department of Artificial Intelligence and Data Science for his guidance and encouragement throughout the project work. I convey my sincere and deepest gratitude to our internal guide, **Dr. E K Subramanian,** Associate Professor, Department of Artificial Intelligence and Data Science, Rajalakshmi Engineering College for her valuable guidance throughout the course of the project. Finally, I express my gratitude to my parents and classmates for their moral support and valuable suggestions during the project.

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ABSTRACT

This Principles of Data Science mini project studies plant species identification from vegetation plot images containing multiple plants. The task is challenging because several species may appear in one image, plants can overlap, and visual differences between species may be subtle. The workflow organizes image data, prepares labels and inputs, applies a deep-learning image representation/classification approach, and explores Test-Time Augmentation (TTA) by evaluating transformed versions of an image and combining prediction outputs. The project workflow includes data inspection, preprocessing, model inference, augmentation-based prediction, and qualitative or metric-based evaluation where verified outputs are available. The report describes the methodology and implementation without claiming unverified accuracy values or experimental improvements. The approach is relevant to biodiversity observation, ecological monitoring, and automated plant image analysis.

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CHAPTER 1

INTRODUCTION

Plant diversity monitoring relies on identifying species from observations collected in natural environments. Vegetation plot images provide a record of plants within a defined area, but a single image may contain many species, partially hidden leaves, different growth stages, and varying illumination. Manually identifying every visible species is time-consuming and can be inconsistent. Computer vision and deep learning can help extract useful visual features from images, while data-science methods provide a repeatable process for organizing data, generating predictions, and evaluating model behavior. This project explores a plant-identification workflow with particular attention to Test-Time Augmentation.

1.1 Background

The project focuses on high-resolution vegetation plot images in which multiple plant species may be present. Unlike single-plant photographs with a clear background, plot images contain dense vegetation, overlapping leaves, shadows, soil, and plants at different scales. A consistent data pipeline is required to connect image identifiers, labels, preprocessing, model outputs, and evaluation.

1.2 Motivation

Plant identification supports biodiversity surveys, ecological research, habitat assessment, and long-term monitoring. Test-Time Augmentation is considered because predictions may change when an image is viewed under simple transformations such as flips or rotations.

1.3 Objectives

The objectives are to understand the plant-image identification problem; inspect the available image and label files; prepare a reproducible preprocessing workflow; apply a deep-learning model to obtain species predictions; investigate Test-Time Augmentation by combining predictions from transformed inputs; and document the evaluation process. The project also aims to identify practical challenges such as class imbalance, multiple species in one image, visual similarity between species, and computational limits. Only verified outputs should be reported as experimental results.

1.4 Problem Statement

Identifying multiple plant species from vegetation plot images is difficult because different species can share similar visual features, plants may overlap, and image conditions vary. The project therefore studies a data-science workflow that processes vegetation images and produces candidate species predictions using deep learning, while examining whether Test-Time Augmentation can make predictions less sensitive to image orientation or other simple visual changes.

1.5 Scope of the Project

The scope includes image and label inspection, image preprocessing, model-based species prediction, Test-Time Augmentation, and evaluation using available verified outputs. The project is an academic prototype intended to study the methodology rather than replace botanical expertise. Predictions should be treated as suggestions requiring validation, especially for rare species and images containing many overlapping plants.

CHAPTER 2

LITERATURE SURVEY (EXISTING SYSTEM)

Plant identification from images has been studied using handcrafted visual descriptors and machine-learning methods, followed by convolutional neural networks and vision-transformer architectures. Deep models can learn image representations from large collections of labelled examples, but their performance depends on the training data, label quality, class coverage, and similarity between training and evaluation images. Vegetation plot imagery introduces additional complexity because a photograph may contain several species rather than one centered plant.

Image classification pipelines commonly resize or normalize images before passing them to a trained model. Convolutional neural networks learn local visual patterns through convolutional filters, while vision transformers use attention to model relationships between image patches. Transfer learning can make pretrained visual representations useful when the available labelled dataset is limited. In this project, the model and input pipeline are treated as separate components so that preprocessing and inference can be inspected independently.

Test-Time Augmentation applies multiple label-preserving transformations to an input image at inference time. Examples include horizontal or vertical flips and rotations where those transformations are appropriate for the task. Each transformed image is passed through the model, and the resulting prediction scores can be combined, for example by averaging. TTA does not train the model or guarantee better results; its value must be assessed against predictions from the original image using the same evaluation data.

The proposed study brings together image-data preparation, deep-learning inference, and TTA in a single repeatable workflow. The baseline prediction is generated from the original image, while augmented predictions are obtained from transformed versions of the same image. Comparing these outputs helps examine prediction stability. Evaluation should use consistent labels, a clearly defined split, and suitable multi-label measures when each image can contain multiple species.

CHAPTER 3

ARCHITECTURE OF THE PLANT IDENTIFICATION SYSTEM

The architecture follows an image-processing pipeline. Vegetation plot images and their associated labels form the input. A data-inspection and preprocessing stage checks image files, identifiers, dimensions, and label format. A pretrained or trained deep-learning model produces species scores for the original image. The TTA stage applies selected transformations, runs inference on each transformed image, maps outputs to a consistent class order, and combines the scores. The final stage stores predictions and evaluates them against verified labels where available.

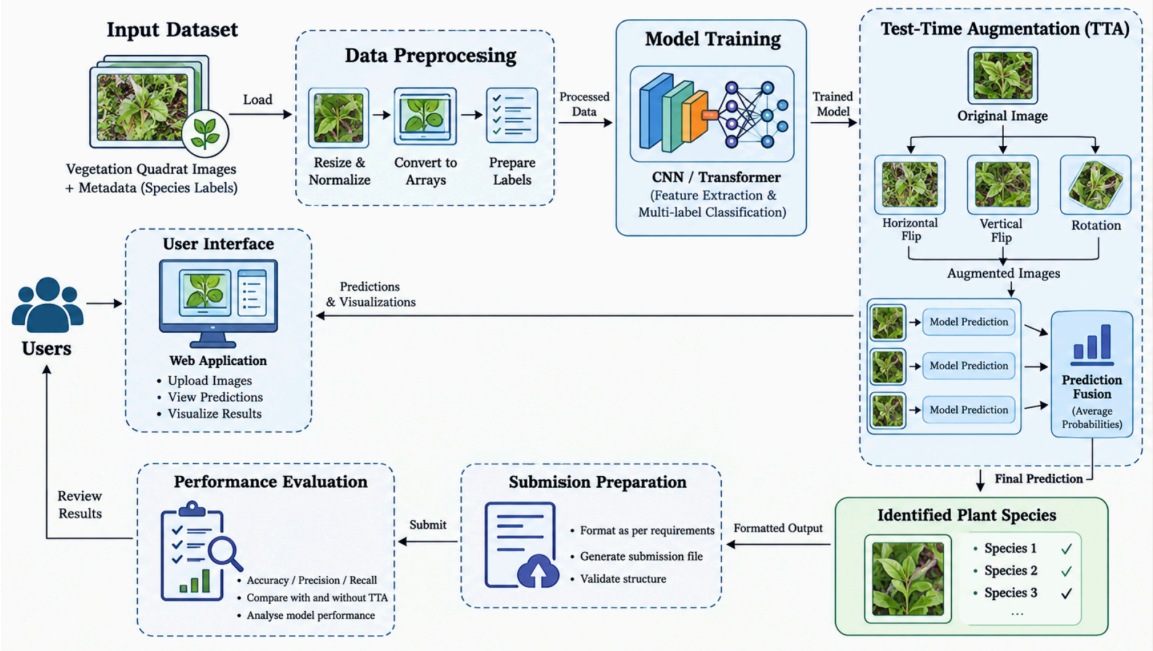


Figure 1: Architecture of the Plant Species Identification Workflow

Input Data Layer

The input consists of vegetation plot images and available metadata or species-label files. Each image must be associated with the correct identifier

and label information. Dataset inspection should identify missing files, duplicate identifiers, malformed labels, and differences in image dimensions. Exact image counts and class totals should be taken from the dataset actually used in the implementation rather than estimated.

Preprocessing Layer

The preprocessing stage loads images, converts them into the format expected by the selected model, resizes them to the required input dimensions, and applies the model-specific normalization. Label parsing must preserve the relationship between each image and its species labels. Any filtering or split creation should be recorded so that evaluation can be reproduced. Training-only transformations must be kept separate from inference-time transformations.

Deep-Learning Inference Layer

The baseline model receives the preprocessed original image and produces a vector of scores for the supported species classes. The model may use a convolutional or transformer-based visual representation, depending on the implementation selected for the project. The class-index mapping must remain consistent between the model output, species identifiers, and any final prediction file. Model architecture and pretrained weights should be documented from the actual code used.

Test-Time Augmentation Layer

TTA creates transformed views of an input image, such as horizontal flips or rotations, and sends each view through the same model. The prediction vectors are aligned to the same class order and combined using a defined rule,

such as averaging scores. TTA is an inference strategy and does not update model weights. The selected transformations must preserve the task meaning and should be compared with a baseline run on the original image.

Prediction and Evaluation Layer

The output stage ranks candidate species or produces multi-label predictions according to the project's label format. Where ground-truth labels and a valid evaluation split are available, baseline and TTA outputs can be compared using suitable metrics such as precision, recall, F1-score, or mean average precision, depending on the evaluation protocol. If verified metrics are unavailable, the report should describe the evaluation procedure without inserting numerical results.

CHAPTER 4

SYSTEM ANALYSIS

4.1 Existing System

Traditional plant identification depends on manual inspection, botanical field guides, and expert knowledge. These methods remain important but can be time-consuming when image collections are large. Simple image classifiers designed for one centered plant may not represent the complexity of vegetation plots with overlapping plants and several species. Inconsistent image preprocessing or label mapping can also make automated results difficult to reproduce.

4.2 Proposed System

The proposed system uses a structured data-science workflow for vegetation plot images. It inspects image and label files, applies consistent preprocessing, runs a deep-learning model to obtain species scores, and optionally combines predictions from transformed views using Test-Time Augmentation.

4.3 System Requirements

Hardware requirements include a computer with sufficient memory and storage for image files and model weights; a compatible GPU can accelerate inference but is not assumed to be available. Functional requirements include loading images and labels, preprocessing inputs, generating model predictions, applying optional TTA, and saving outputs. Non-functional requirements include reproducibility, clear class mapping, efficient memory use, and honest reporting of evaluation limitations.

CHAPTER 5

MODULES DESCRIPTION

5.1 Dataset Inspection Module

The Dataset Inspection Module checks the available image files and metadata before model execution. It examines identifiers, label fields, file readability, image dimensions, missing values, and possible duplicates. The output of this stage should describe the actual dataset used. Dataset sizes and class frequencies are not inserted here unless they have been measured from the working files.

5.2 Image Preprocessing Module

The Image Preprocessing Module prepares each image for the selected model. Typical operations include decoding, resizing to the model input dimensions, converting channel order if required, and applying the model's expected normalization. Preprocessing must be identical for the baseline and TTA branches apart from the intended augmentation transformation, so that comparisons remain meaningful.

5.3 Baseline Prediction Module

The Baseline Prediction Module passes the original preprocessed image through the selected deep-learning model and records the output scores. The scores are mapped to the corresponding species classes and sorted to obtain candidate predictions. This baseline establishes the reference against which the TTA output can be compared. The report does not assume a particular score or ranking without verified model output.

5.4 Test-Time Augmentation Module

The Test-Time Augmentation Module generates selected transformed versions of each input image, performs inference for each view, and combines the score vectors. Common transformations include horizontal flips and rotations when suitable for the image task. The aggregation rule and transformation set should be kept consistent across evaluation runs. TTA can increase inference cost and does not guarantee improved classification quality.

5.5 Prediction Evaluation Module

The evaluation module compares model predictions with the available reference labels using a metric suited to the label structure. For multi-label images, metrics may include precision, recall, F1-score, or mean average precision if the required ground truth and evaluation protocol are available. Baseline and TTA runs must use the same evaluation examples. Only metrics produced by an actual completed run should be included in the final results.

5.6 Output and Visualization Module

The output module saves ranked species predictions and any supporting summaries in a reusable format such as CSV. Visual summaries may compare baseline and TTA predictions, show the transformations used, or display sample predictions for manual review. Screenshots and numerical charts should be generated from the running implementation; illustrative graphics must not be presented as measured experimental results.

CHAPTER 6

IMPLEMENTATION

The implementation begins by loading the image metadata and verifying that the referenced image files are available. The preprocessing pipeline converts each image to the expected dimensions and normalization before inference. File paths, class identifiers, and output columns should be checked to prevent mismatches. If the dataset is too large to load into memory at once, images should be processed in batches.

The deep-learning component runs inference on the original image and produces species scores. Because TTA multiplies the number of forward passes, batch size and image resolution should be chosen with the available hardware memory in mind. A baseline-only mode is useful for debugging and comparison.

The evaluation stage reads the reference labels and prediction outputs, validates their formats, and computes the selected metrics when the necessary ground truth is available. The workflow should log skipped or unreadable images and avoid silently treating missing labels as negative labels. Any model checkpoints or preprocessing settings used for a run should be documented.

The implementation is organized into separate components for input validation, preprocessing, baseline inference, TTA inference, and output evaluation. This modular design allows each stage to be checked independently and makes it easier to identify errors in file paths, labels, image shapes, or class ordering. Computational limitations and unverified steps should be clearly documented in the final academic submission.

CHAPTER 7

RESULTS AND DISCUSSION

The project establishes a workflow for processing vegetation plot images and generating candidate plant-species predictions. The data-inspection stage is intended to verify image availability, label consistency, and class mapping before inference. The baseline and TTA branches use the same model and class ordering, with transformations applied only in the TTA branch. Actual dataset counts and evaluation metrics should be inserted only after they have been confirmed from the completed run.

A fair comparison between baseline inference and TTA requires both methods to be evaluated on the same images with the same reference labels and metric definitions. TTA may improve stability for some images, have little effect for others, or reduce quality if transformations are unsuitable. No numerical improvement is claimed in this report without verified evaluation output. Qualitative inspection can help identify difficult examples, visually similar species, occlusion, and errors caused by background or lighting.

The output stage can present ranked species predictions and compare the score distribution from the original image with the aggregated TTA scores. A schematic diagram is not evidence of measured model performance.

Overall, the project demonstrates how image preprocessing, deep-learning inference, and test-time augmentation can be combined into a repeatable data-science workflow for plant identification. Its usefulness depends on the quality and coverage of the labels, the visual diversity of the dataset, the selected model, and the validity of the evaluation protocol. Results should be interpreted as model predictions that may require expert confirmation.

CHAPTER 8

FUTURE ENHANCEMENTS

The project can be extended by evaluating additional model architectures, refining image preprocessing, and comparing alternative TTA transformations and score-aggregation methods. A clearer evaluation protocol with verified ground truth can support reliable comparison of baseline and augmented inference. Class imbalance, rare species, and multi-label prediction can be studied in greater detail. Model explanations and confidence calibration may also help users interpret predictions responsibly.

Future work may include testing more robust vision models, selecting augmentations based on plant-image characteristics, and measuring the computational cost of each inference strategy. The workflow could be extended with error analysis by species, reproducible configuration files, and a simple interface for reviewing candidate predictions. Additional validation across locations, seasons, lighting conditions, and image sources would be needed before considering practical ecological use.

CHAPTER 9

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